

SCHOOL OF ENGINEERING, CUSAT
B. TECH DEGREE 1ST SEMESTER SECOND INTERNAL EXAMINATIONS,

JULY 2023

23-200-0201A COMPUTER PROGRAMMING AND PROBLEM SOLVING (ME-A)

(2023 Scheme)

TIME: 2 hrs

MAX. MARKS: 30

Course Outcomes

On completion of this course the student will be able to:

CO1	<i>Elucidate the basic architecture and functionalities of a computer and also recognize the hardware parts.</i>
CO2	<i>Apply programming constructs of C language to solve real-world problems.</i>
CO3	<i>Explore user-defined data structures like arrays in implementing solutions to problems like searching and sorting</i>
CO4	<i>Explore user-defined data structures like structures, unions and pointers in implementing solutions</i>
CO5	<i>Design and Develop Solutions to problems using modular programming constructs using functions</i>

PART A

(Answer all questions)

(5 x 2= 10 Marks)

Q. No.	Questions	Marks	BL	CO	PO
I (a)	Explain the declaration of one dimensional and two-dimensional arrays with an example.	2	L1	4	1-3
(b)	Differentiate between call by value and call by reference.	2	L1	5	1-3
(c)	Write a C program for find the length of string without using in-built functions	2	L2	4	1-3
(d)	What is function? Explain the difference between user defined and library functions	2	L1	5	1-3,5
(e)	Predict the output: <pre>#include <stdio.h> #include <string.h> int main() { char str[] = "Hello, world!"; int length = strlen(str); printf("The length of the string \"%s\" is %d.\n", str, length); return 0; }</pre>	2	L2	4	3

PART B
(Answer the following questions
choosing either the first OR second option)

(1 X 10= 10 Marks)

Q.No.	Questions	Marks	BL	CO	PO
II (a)	Write a C program to read N integers into an array A and to find the (i)sum of odd numbers (ii) sum of even numbers (iii) Average of all numbers	10	L3	4	1-3
OR					
III (a)	Write a C program to search an element using binary search.	10	L3	4	1-3
(1 X 10= 10 Marks)					
IV (a)	Explain the string handling functions in C	6	L3	5	1-3,5
(b)	Write a C program to swap two numbers using function.	4	L3	5	1-3,5
OR					
V (a)	Print mark sheet of 'n' students using structure	6	L3	5	1-3,5
(b)	Differentiate between structure and union	4	L3	5	1-3,5

Bloom's Taxonomy Levels

L1 - %	L2 - %	L3-%	L4-%	L5-%	L6-%
23.07	15.38	61.53	0	0	0